

■ TECHNICAL REPORT: AI CAREER ACCELERATOR & SKILL GAP PLATFORM

Full-Stack Architecture • Skill Gap Analytics Engine • Interactive Visualizations • Public Cloud Deployment

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1. Executive Summary & Abstract

The **AI Career Accelerator & Skill Gap Platform** is an enterprise-grade technical web application engineered to quantify student job readiness, analyze resume ATS quality, categorize technical skill gaps, and generate dynamic learning roadmaps for competitive engineering roles (Data Scientist, ML Engineer, Data Analyst, Full-Stack AI Engineer). Built entirely in Python using Streamlit, Plotly, Pandas, PyPDF, and Scikit-Learn, the system delivers an end-to-end analytical pipeline operating under 200ms latency.

2. System Architecture & Modular Design

The system follows a **4-Tier Decoupled Architecture** separating presentation, analytics engines, taxonomy definitions, and cloud deployment pipelines:

Layer Tier	Components & Modules	Technical Responsibilities
1. Presentation Layer	Streamlit UI, Custom CSS, Plotly Charts	Interactive rendering, sidebar navigation, reactive widgets.
2. Analytics Engine	resume_analyzer, skill_gap_engine, roadmap_generator, project_recommender, scoring_algorithm, dashboard	PDF text parsing, project coverage, weighted scoring, interview questions, taxonomy mapping.
3. Data & Taxonomy	sample_data, Canonical Skill Taxonomy, Question Bank, Rubric	Region-Specific backup dictionaries, related skill ontology graphs.
4. Infrastructure	Git, GitHub, Streamlit Community Cloud	Continuous Deployment triggers, serverless container hosting.

3. Core Technical Modules & Analytical Algorithms

3.1 Skill Extraction & Word Boundary Regex Matching

To eliminate false positive matches (e.g., matching 'R' inside 'REACT' or 'Java' in 'JavaScript'), skill extraction employs regular expression word boundaries `\b` mapped against a 50+ canonical taxonomy:

```
pattern = r'\b' + re.escape(term) + r'\b'
if re.search(pattern, lower_text):
    found_skills.add(canonical_name)
```

3.2 ATS Resume Quality Scoring Formula

The ATS Score (0-100) evaluates 5 structural criteria: Contact Details (15 pts), Core Sections (20 pts), Impact Metrics & Action Verbs (25 pts using regex `(\d+%|\d+k|\d+|\b|\$|\d+)`), Keyword Density (25 pts), and Word Count (15 pts).

3.3 Weighted Job Readiness Calculation

Job readiness is computed by weighting exact strong skill matches (1.0 weight) and foundational moderate skill matches (0.5 weight):

Readiness Score (%) = min(100.0, [Count(Strong)*1.0 + Count(Moderate)*0.5] / TotalRequired * 100)

3.4 Phased Learning Roadmap & Matrix Project Matcher

The roadmap engine dynamically organizes target gap skills into 5 structured phases (Foundations → Frameworks → Production MLOps → Capstone Projects → Interview Prep). The project recommender ranks project specifications using match score: $Score = 3 * GapMatches + 2 * RoleMatch$.

4. Performance Benchmarks & Validation Results

Test Candidate Profile	Target Job Role	Readiness %	ATS Score	Identified Key Missing Skills
Alex Rivera (Data Science)	Data Scientist	51.9%	92 / 100	PySpark, MLOps, A/B Testing, Docker
Alex Rivera (Data Science)	Machine Learning Engineer	42.8%	92 / 100	Kubernetes, CUDA, MLflow, CI/CD
Sam Chen (Software Dev)	Junior Data Analyst	36.3%	75 / 100	Power BI, Tableau, Advanced Statistics

5. Tech Stack Summary & Deployment

Frontend: Streamlit 1.62.0 | **Charts:** Plotly 6.9.0 | **Data:** Pandas & NumPy | **PDF Ingestion:** PyPDF 6.16.1 | **Deployment:** Streamlit Community Cloud (Public Serverless Container)
Public Live App URL: <https://ai-career-navigator.streamlit.app>
GitHub Repository: <https://github.com/likithyadav128-tech/ai-career-navigator>